PROJECT REPORT
PayProof

B2B Order Evidence & Reconciliation Platform



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DA-3

Project / Product	**PayProof**
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1. Abstract

PayProof is a business-to-business (B2B) order evidence and reconciliation application intended to organize order records, supporting documents, lifecycle progression, evidence completeness, reconciliation exceptions, and payment disputes in a single workspace. The supplied application screenshots show an order dashboard, order creation, evidence upload, status controls, evidence-pack ZIP export, dispute handling, an audit-ledger view, an evidence-integrity information page, and a Cognito sign-in interface.

Project notes describe a test deployment on Amazon Web Services (AWS) using a CloudFront-hosted frontend and services including API Gateway, Lambda, Cognito, Amazon S3, AWS KMS, DynamoDB, and CloudWatch. The report presents these as the reported deployment design; current resource state and security settings must be confirmed with the AWS Console screenshots reserved in Section 8.

The appropriate current assessment is: MVP demonstrated; controlled pilot readiness pending verification. Screenshots demonstrate visible interface states but do not, by themselves, establish data durability, full tenant isolation, malware scanning, audit immutability, cryptographic validation of exported evidence, or production-grade operational controls.

2. Introduction

Business orders commonly involve records from multiple systems: purchase orders, order confirmations, shipping or delivery evidence, invoices, supporting communications, and payment-dispute records. When these materials are separated across email, shared drives, and operational tools, teams can lose time reconstructing the sequence of events, locating missing documents, and preparing a consistent evidence package for review.

PayProof is designed around the order as the primary record. It brings lifecycle information, linked evidence, completeness indicators, reconciliation flags, and dispute history into a single workflow. The intended outcome is a more traceable review process and a consistent export of the evidence associated with an order.

This submission follows the structure of a cloud application project report. It documents the visible MVP, the AWS architecture described in project notes, the checks reported to date, and the infrastructure evidence that should be inserted before final evaluation. Claims not established by the available screenshots are explicitly marked for verification.

3. Objectives and Scope

3.1 Objectives

- Provide a consolidated order workspace with lifecycle stage, value, delivery date, customer entity, and evidence completeness.
- Allow authorized users to create and review order records and associate supporting evidence documents.
- Surface missing-document flags and reconciliation exceptions to guide manual review.
- Provide controlled lifecycle progression and make exceptional transitions visible.
- Support payment-dispute review, correspondence history, and resolution notes.
- Provide an evidence-pack ZIP export and a chronological audit-ledger interface.
- Use AWS-managed application, identity, database, and storage services for the reported test deployment.

3.2 Scope

This report covers the application behavior visible in the supplied screenshots and deployment details present in the project notes. It does not claim that a visible button proves the complete underlying service path is correct. Backend persistence, tenant-boundary enforcement, scan verdicts, authorization of overrides, export-manifest validation, key policies, backup recovery, and operational alarms require direct test results or AWS Console evidence.

4. System Architecture

The reported architecture separates the browser-facing frontend from backend APIs and managed data services. CloudFront serves the web interface; API Gateway provides the API entry point; Lambda functions implement backend request handling; Cognito is the reported authentication service; DynamoDB stores application records; S3 stores frontend assets and uploaded evidence; KMS is referenced for encryption-key management; and CloudWatch supports logging and monitoring. The exact route-to-function mapping and resource policies should be confirmed in the AWS Console and infrastructure template.

Layer	AWS component	Role / verification
Presentation layer	Browser / CloudFront	Displays dashboard, orders, evidence controls, disputes, audit ledger, and sign-in. Verify distribution status, TLS, origins, and cache behavior.
API layer	Amazon API Gateway	Exposes API routes for orders, disputes, context, health, and audit information as reported. Confirm routes, authorization, CORS, and throttling.
Compute layer	AWS Lambda	Runs backend handlers. Confirm function names, runtime, environment configuration, execution role, logs, and error handling.
Identity layer	Amazon Cognito	Provides the reported sign-in / token flow. Confirm app-client settings, callback URLs, token duration, and demo access controls.
Data layer	DynamoDB	Stores order, evidence metadata, dispute, and audit-related records as implemented. Confirm table schema, encryption, backup, and access enforcement.
Object / key layer	Amazon S3 and AWS KMS	S3 stores frontend files and evidence objects; KMS is referenced for encryption. Confirm bucket privacy, object authorization, key policy, and actual encryption use.
Operations layer	Amazon CloudWatch	Supports backend logs and monitoring. Confirm log retention, useful metrics, alarms, redaction, and access policies.

4.1 High-level request flow

User browser -> CloudFront frontend -> API Gateway -> Lambda handler -> Cognito identity/authorization context and managed data services (DynamoDB, S3, and KMS where configured) -> CloudWatch logs and monitoring. This is a conceptual flow based on the reported service set, not a substitute for the deployed architecture diagram or resource-level screenshots.

5. AWS Services Used

AWS service	Reported resource	Purpose / evidence needed
Amazon CloudFront	Reported frontend delivery endpoint	Serves the web application through a distribution. Validate distribution status, domain, TLS, origins, and deployment timestamp.
Amazon API Gateway	Reported API endpoint in ap-south-1	Publishes application API routes. Validate stage, routes, authorizers, CORS, and throttling.
AWS Lambda	Backend request handlers	Processes API operations. Validate function list, runtime, roles, recent invocations, errors, and configuration.
Amazon Cognito	User Pool ap-south-1_ZfV6Q3TX2; app client 6ib5a4n4u5e2r1oppk8esb8pc4	Supports reported sign-in interface. Do not treat identifiers alone as proof of secure configuration.
Amazon S3	Frontend bucket and evidence bucket reported	Stores frontend assets and evidence objects. Verify Block Public Access, encryption, policies, versioning/retention, and object-level authorization.
AWS KMS	Encryption key service referenced in notes	Manages cryptographic keys where configured. Verify key status, key policy, grants, rotation, and service usage.
Amazon DynamoDB	payproof-records-test	Stores application records. Verify key schema, encryption, indexes, point-in-time recovery/backups, and tenant scoping.
Amazon CloudWatch	Backend logs and monitoring	Records application events and errors. Verify log groups, retention, alarms, and safe logging practices.
AWS SAM / CloudFormation	Stack payproof-test reported	Supports deployment management. Capture stack status, region, change history, and outputs.

Reported test environment: AWS Region ap-south-1; CloudFormation/SAM stack payproof-test. Reported CloudFront URL: https://d10pwugd4h3vow.cloudfront.net. These values were supplied in the project notes and must be rechecked before submission or operational use.

6. Application Features

6.1 Order dashboard

Shows active-order count, evidence-completeness percentage, blocked-invoice count, active disputes, and the order list. Rows show order reference, customer entity, lifecycle stage, value, completeness, reconciliation status, delivery date, and forensic actions.

6.2 Order creation

The Create Order Record dialog captures order reference, expected delivery date, customer legal entity, billing/AP contact, and line-item information such as SKU, description, quantity, and unit price.

6.3 Evidence ingestion

The Ingest Evidence Document dialog includes a document-type selector, optional source reference, and file picker. The detail screen labels the evidence section as SHA-256 hashed. The actual persisted bytes, digest calculation, and download permissions should be tested independently.

6.4 Evidence completeness and reconciliation

The order view shows completeness percentages and flags such as a missing customer purchase order. The score is a useful workflow indicator, but calculation rules should be documented by order type and lifecycle stage.

6.5 Lifecycle control

Lifecycle controls are visible in order details. A screenshot shows an invalid Confirmed-to-Dispatched transition warning and a prompt for administrative override. Server-side enforcement, role authorization, and audit logging for overrides must be verified.

6.6 Evidence-pack export

An Export Verifiable Pack (ZIP) control is visible and the browser downloads panel shows a ZIP download. Archive contents, manifest correctness, included hashes, and tenant isolation are not established by the UI screenshot alone.

6.7 Payment dispute desk

The dispute desk displays a damaged-goods claim, disputed amount, order reference, status, resolution deadline, and correspondence history. A resolution dialog exposes a strategy selector and notes field.

6.8 Audit ledger

The interface presents chronological events with timestamp, actor, role, action, resource target, and summary. The “Immutable” label is a UI label; tamper resistance depends on actual write/delete permissions and storage configuration.

6.9 Evidence-integrity guidance

The information page distinguishes user-entered facts, extracted document values, and system inferences. SHA-256 can detect byte changes relative to a known digest, but by itself does not prove who created a document or whether its contents are legally authentic.

6.10 Authentication

A Cognito sign-in dialog is shown with username/password fields and a demo-login action. Before real customer data is used, rotate any shared demo credentials and disable public demo access unless deliberately restricted and approved.

7. Implementation Details and Demonstration Screenshots

The following screenshots are application-interface evidence supplied with the project. They document visible user journeys and are not intended to replace the AWS Console evidence in Section 8.

7.1 Application dashboard and order list

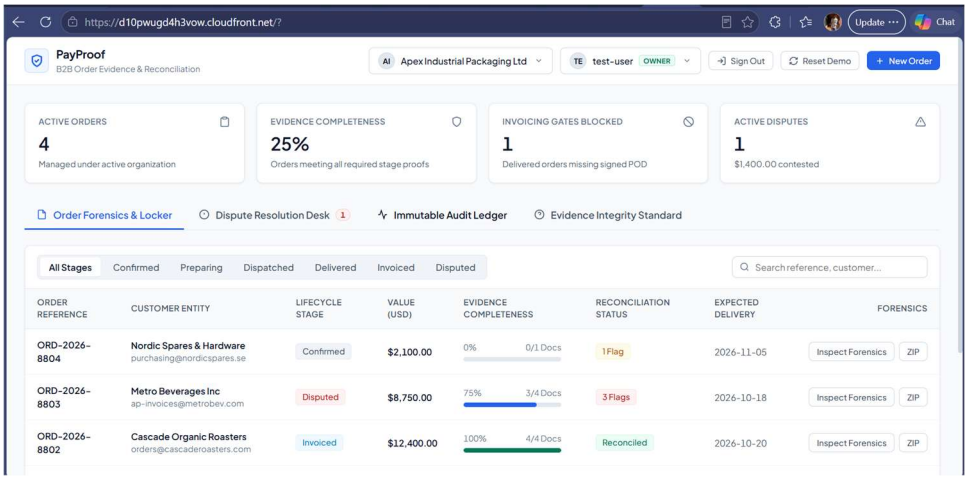


Figure 7.1 - PayProof dashboard with order metrics, evidence completeness, reconciliation status, and recent orders.

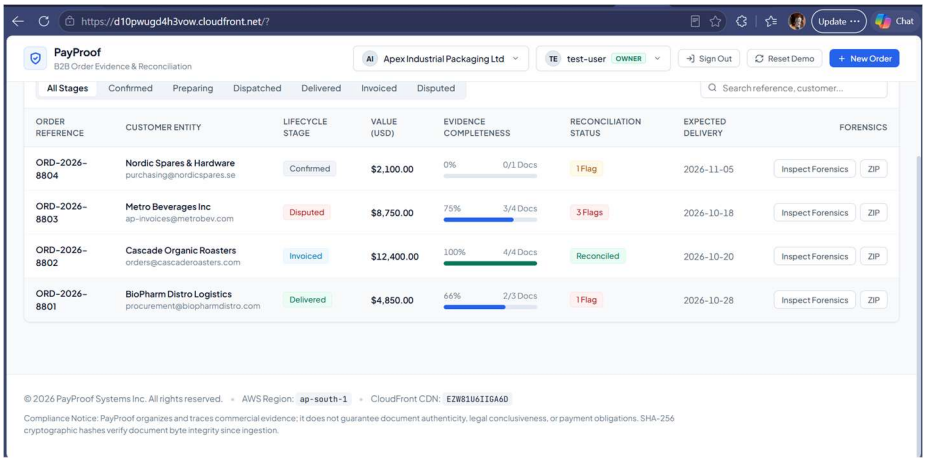


Figure 7.2 - Order list view with customer, lifecycle, value, evidence, and reconciliation columns.

7.2 Order evidence and integrity details

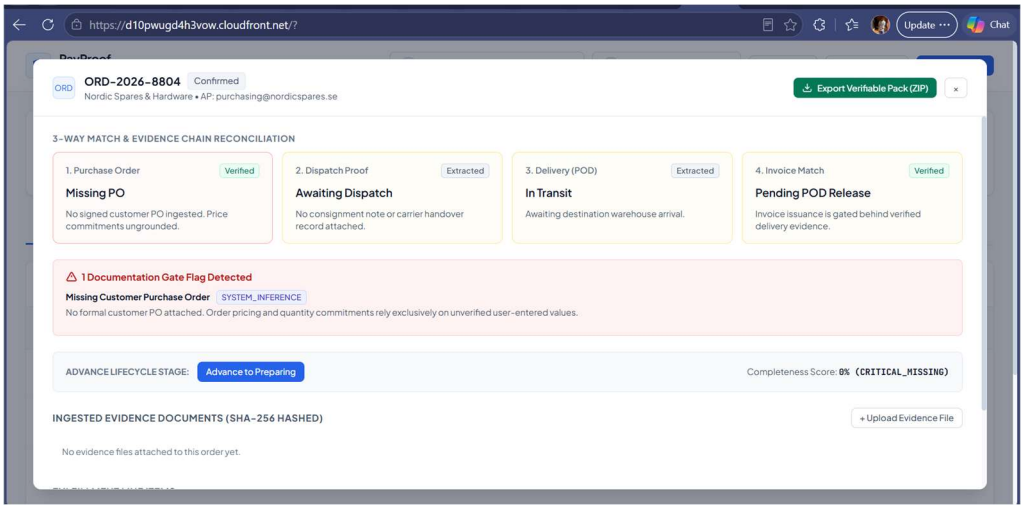


Figure 7.3 - Order detail with lifecycle/evidence summary and missing-document warning.

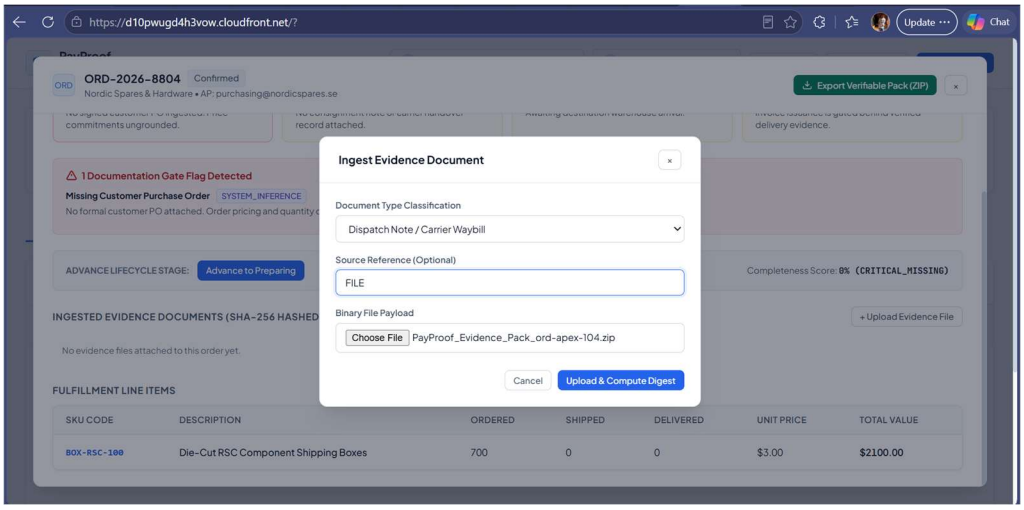


Figure 7.4 - Evidence ingestion dialog showing document classification and file selection fields.

7.3 Disputes and audit history

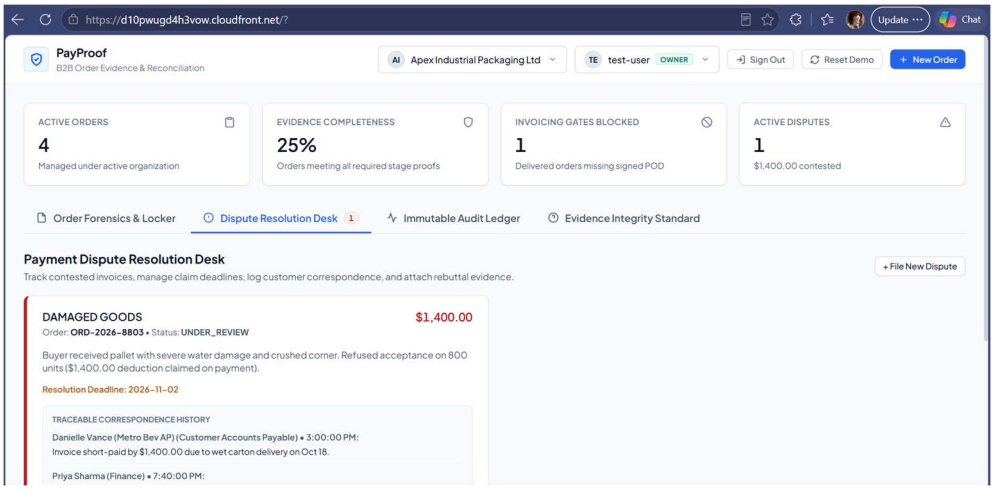


Figure 7.5 - Payment dispute resolution desk showing a dispute record and correspondence history.

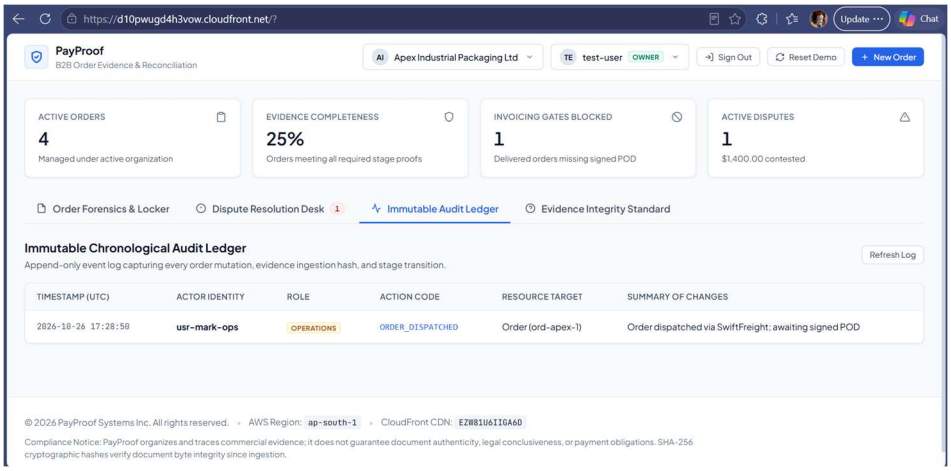


Figure 7.6 - Audit-ledger interface with a chronological recorded event.

7.4 Order creation, evidence standard, and authentication

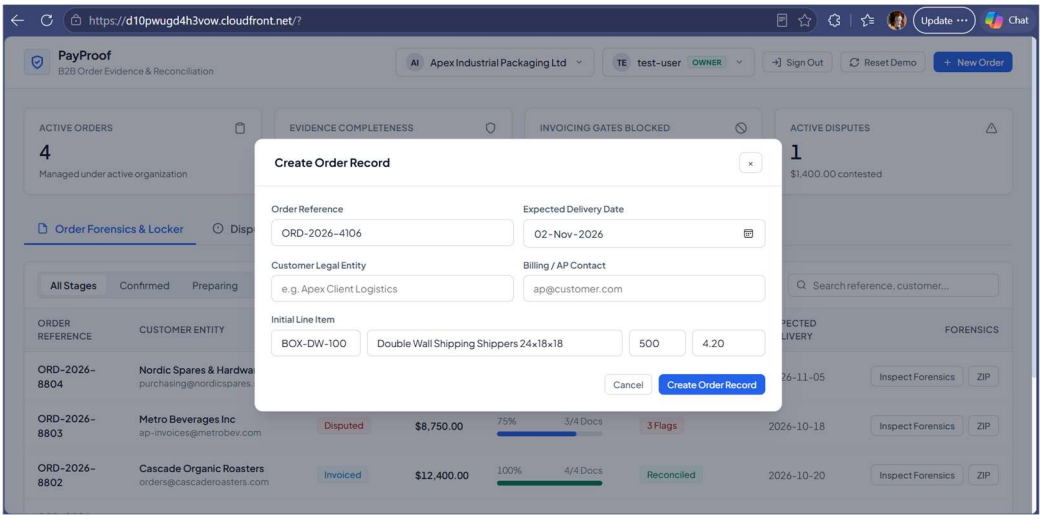


Figure 7.7 - Create Order Record dialog with order, entity, delivery, and line-item fields.

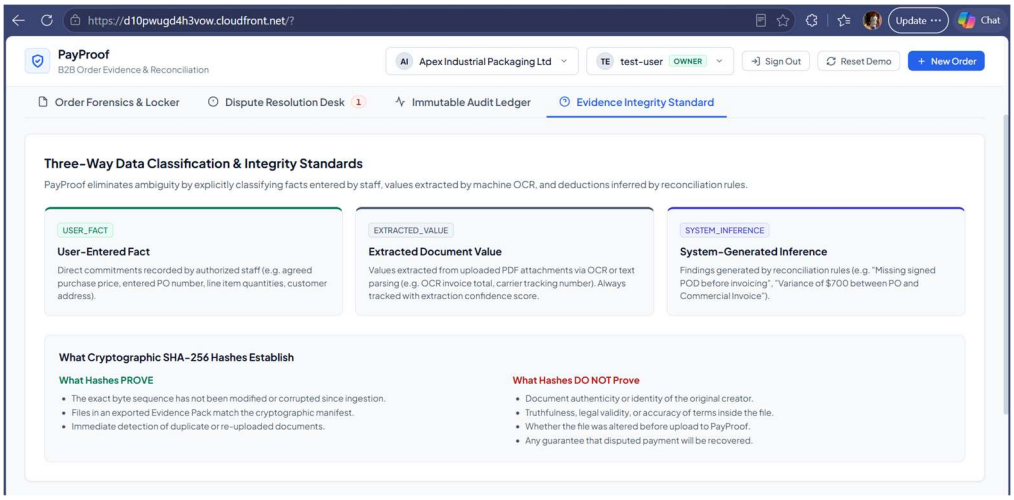


Figure 7.8 - Evidence Integrity Standard page describing evidence classifications and integrity concepts.

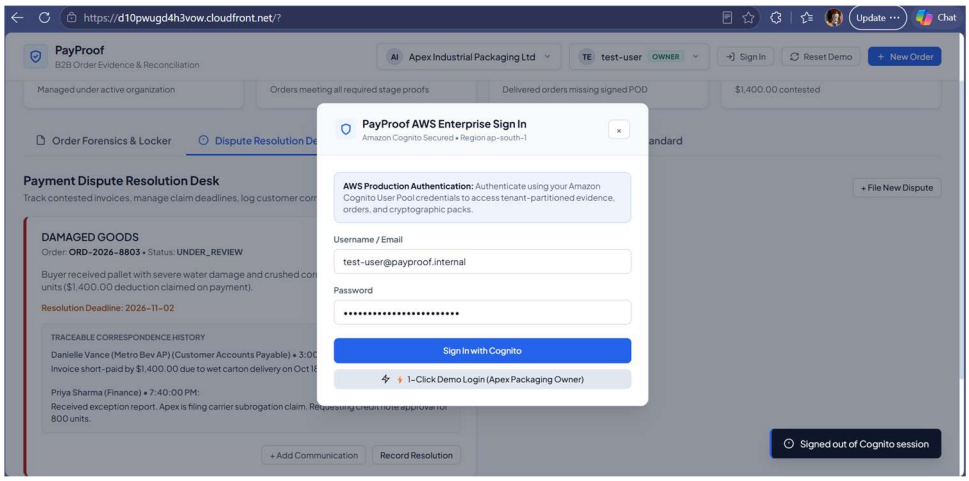


Figure 7.9 - Cognito sign-in interface. Do not include credentials in the submission.

8. AWS Infrastructure Screenshots

The AWS console screenshots below document the resources and selected settings visible in the supplied test account. Each image is evidence of a point-in-time console view. Where the supplied captures do not show a security setting or live test response, the report labels the gap rather than treating the setting as verified.

Resource inventories do not, by themselves, prove that an application is secure or continuously available. Remaining evidence gaps include S3 bucket access/encryption/retention settings, the full scope and trust relationships of IAM roles, CloudWatch Logs retention and alarm actions, and a direct health-check response.

8.1 CloudFormation / SAM stack status

The CloudFormation overview shows the payproof-test stack in UPDATE_COMPLETE in the Asia Pacific (Mumbai) region. Supporting captures show the event history, provisioned resources, and generated outputs. These are point-in-time console views and do not guarantee ongoing availability.

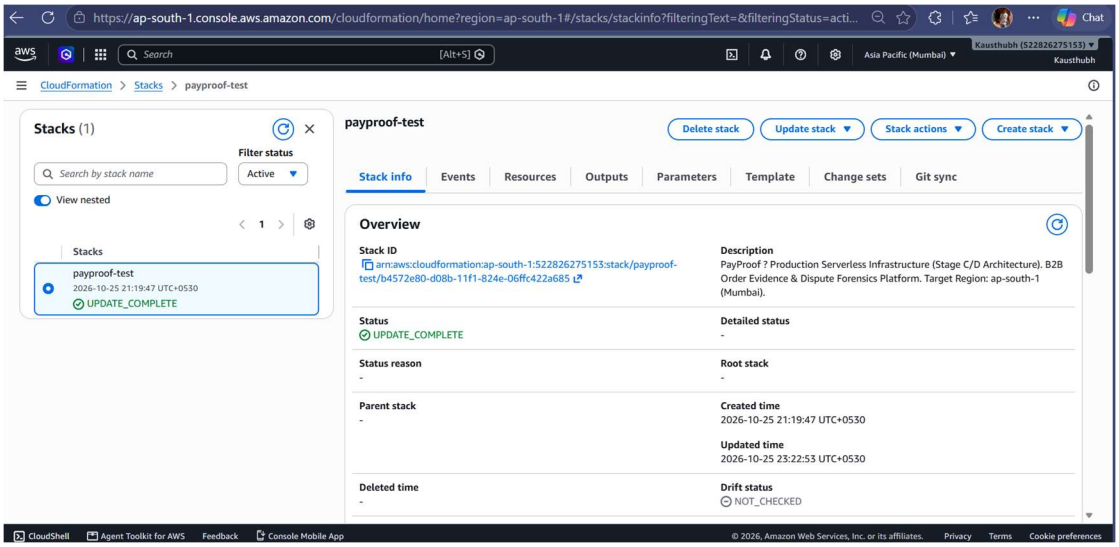


Figure 8.1 - CloudFormation stack payproof-test in UPDATE_COMPLETE state.

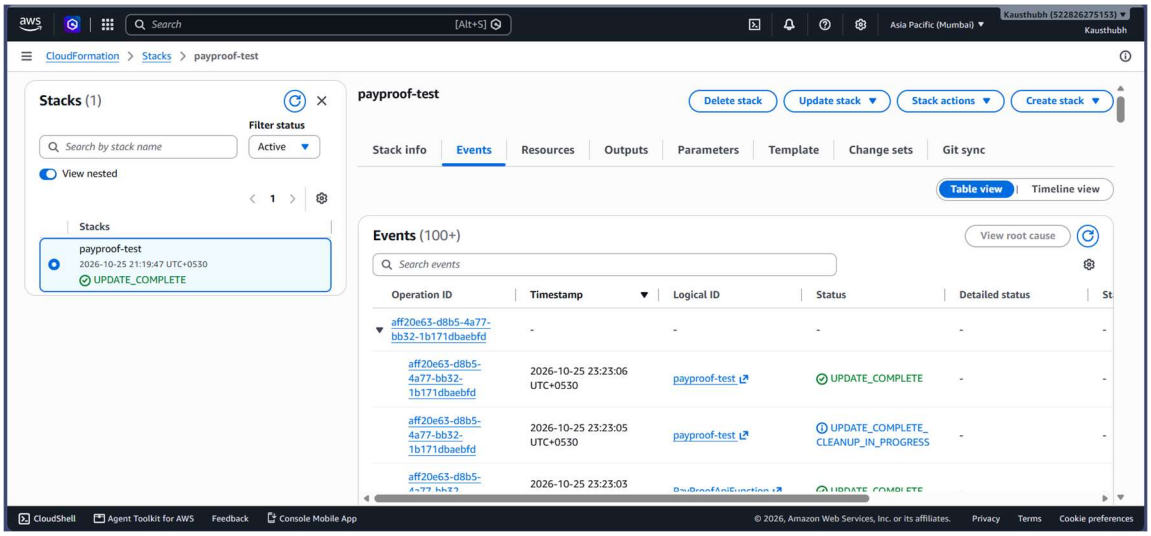


Figure 8.1(a) - CloudFormation event history for the payproof-test stack.

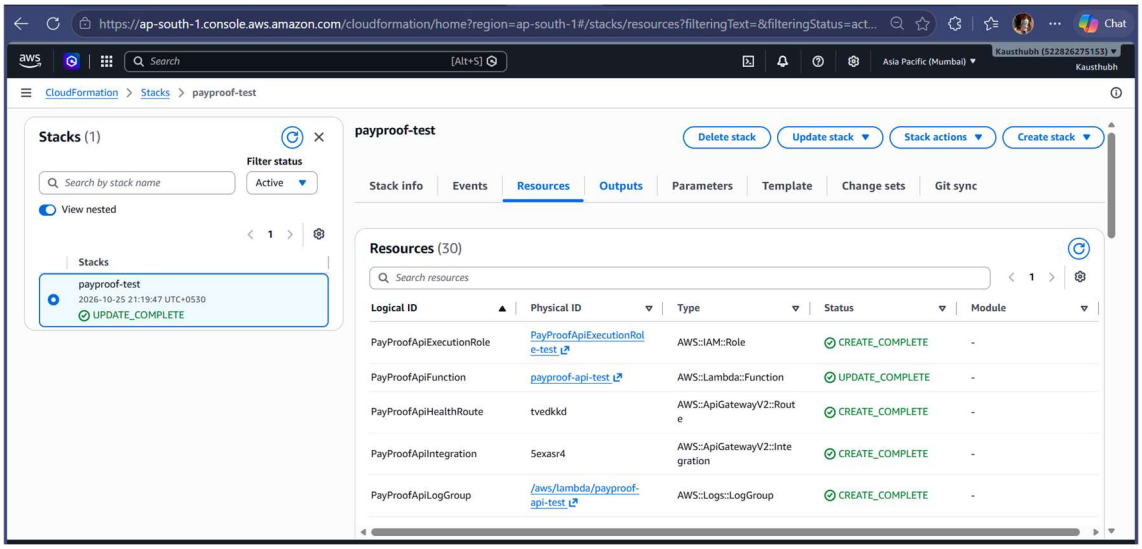


Figure 8.1(b) - CloudFormation resource inventory, including the PayProof API Lambda, API Gateway route, integration, and log group.

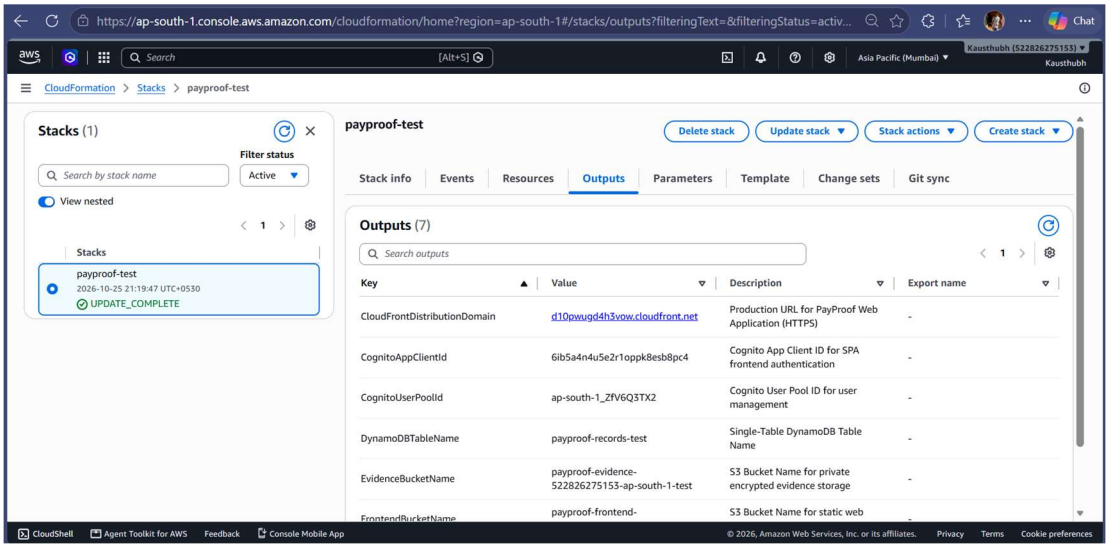


Figure 8.1(c) - CloudFormation outputs listing the frontend domain, Cognito identifiers, DynamoDB table, and S3 buckets.

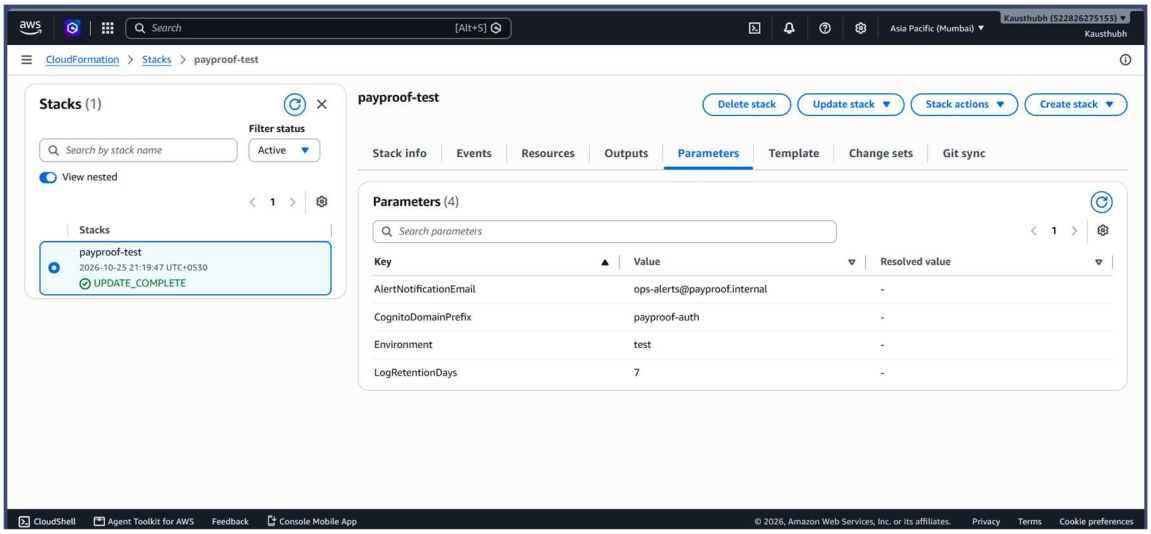


Figure 8.1(d) - CloudFormation parameters for the test environment, Cognito domain prefix, log retention, and alert-notification email.

8.2 CloudFront distribution

The CloudFront overview identifies the PayProof test distribution and its domain. The Behaviors view routes /health, /api/health, and /api/* to API Gateway origins while the default behavior serves the frontend; the displayed API behaviors require HTTPS and the default behavior redirects HTTP to HTTPS. Standard logging is shown as Off and should be reviewed before production use.

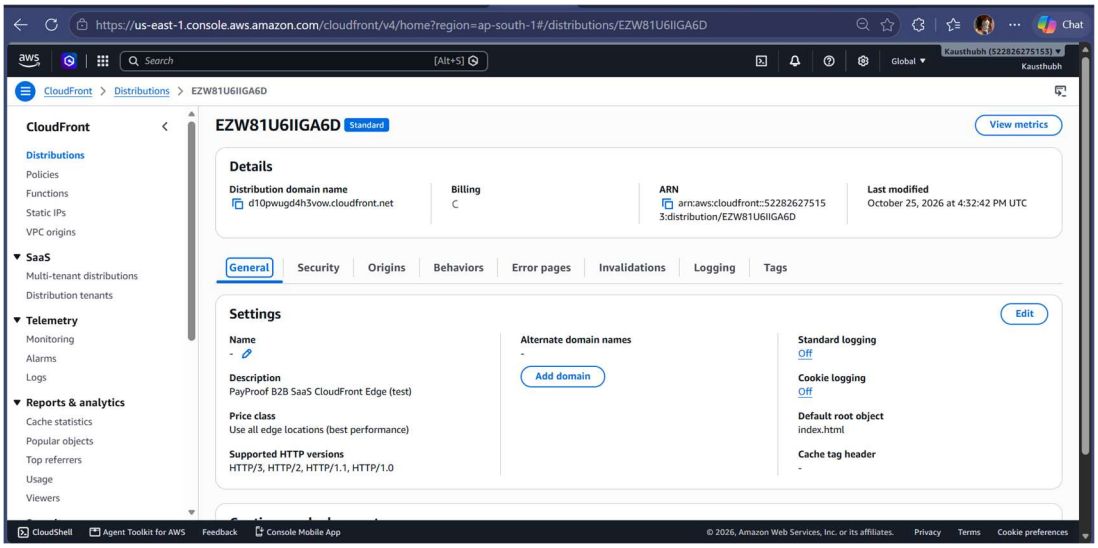


Figure 8.2 - CloudFront distribution details for the PayProof test frontend.

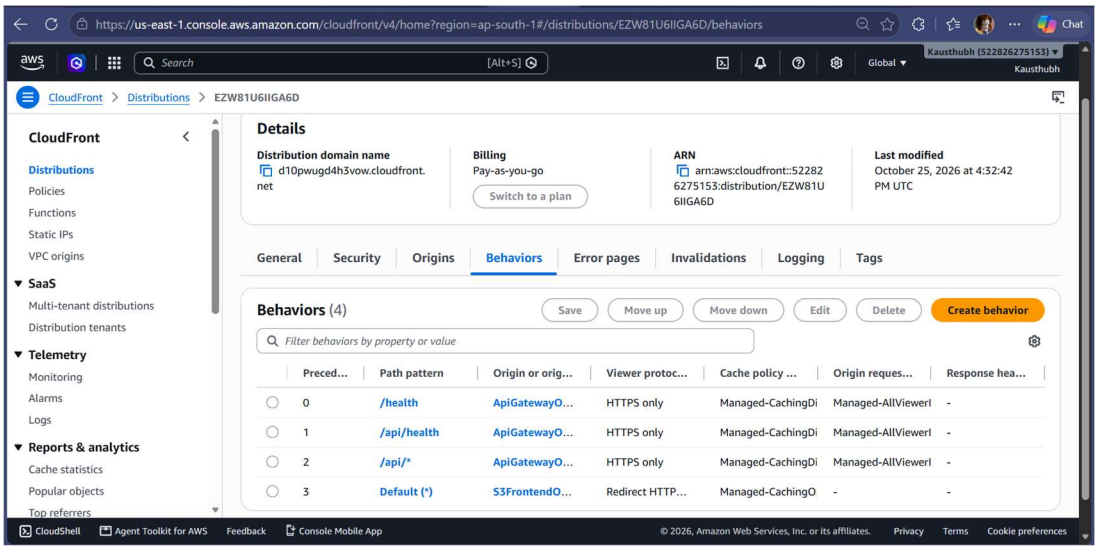


Figure 8.2(a) - CloudFront behavior rules for health and API paths, alongside the default frontend behavior.

8.3 API Gateway HTTP API inventory

The API Gateway inventory lists payproof-http-api-test, an HTTP API with ID yjf4isvot7, in ap-south-1. This list view confirms that the API resource is present. Route-level authorization, deployed-stage settings, CORS, throttling, and request validation are not shown in the supplied image.

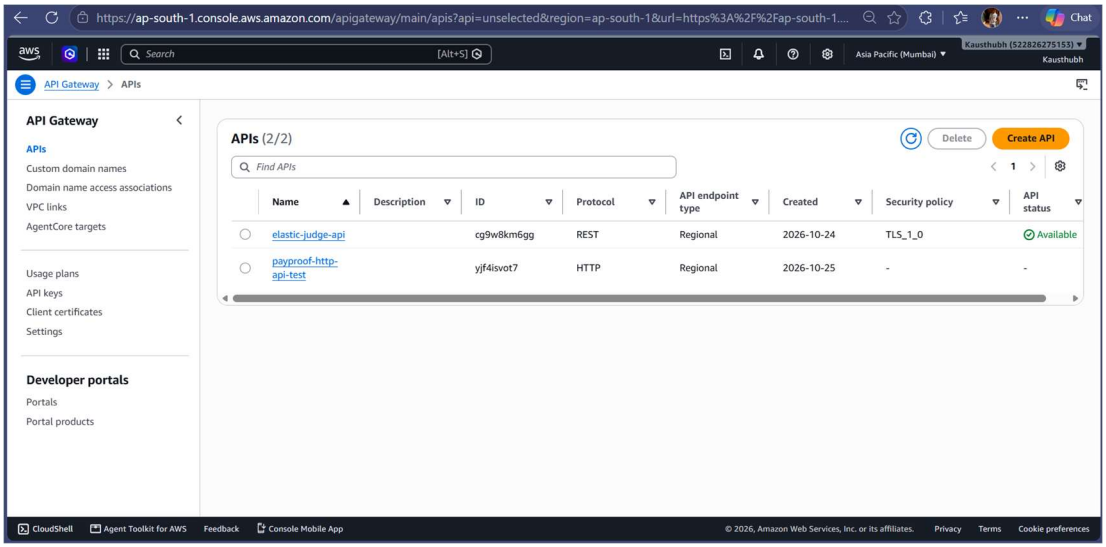


Figure 8.3 - API Gateway inventory showing the PayProof HTTP API.

8.4 Lambda function inventory

The Lambda inventory lists payproof-api-test and payproof-ocr-worker-test, both shown with Python 3.13 runtime. The screenshot confirms function presence and runtime labels only; detailed configuration, triggers, environment variables, and invocation success are not established by this list view.

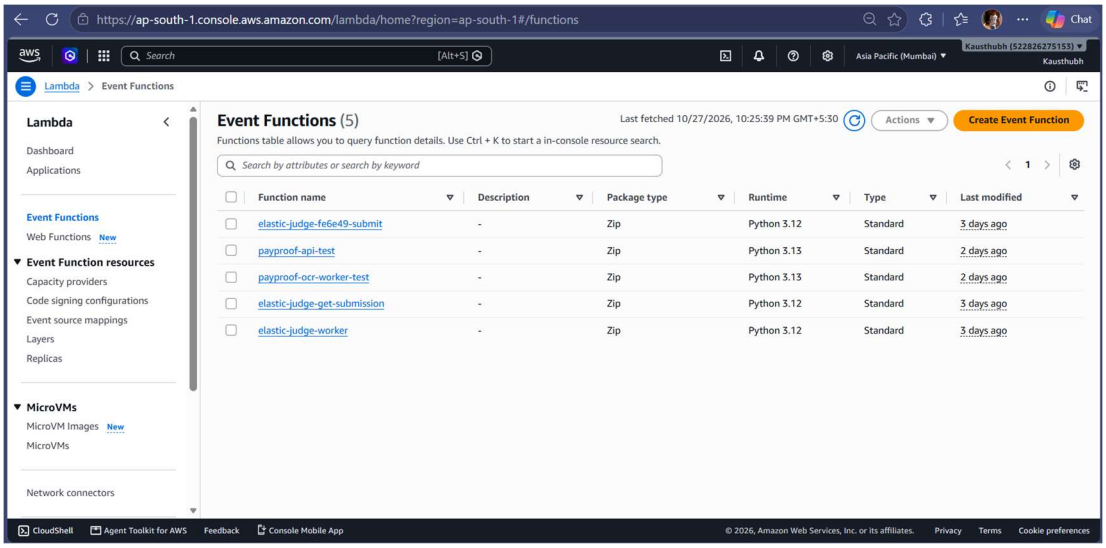


Figure 8.4 - Lambda function inventory listing the PayProof API and OCR worker.

8.5 Cognito User Pool

Amazon Cognito shows the payproof-users-test User Pool with ID ap-south-1_ZfV6Q3TX2. The overview indicates an estimated two users at capture time. App-client authentication flows, callback/logout URLs, token lifetime, and account restrictions are not visible in the supplied views and remain unverified.

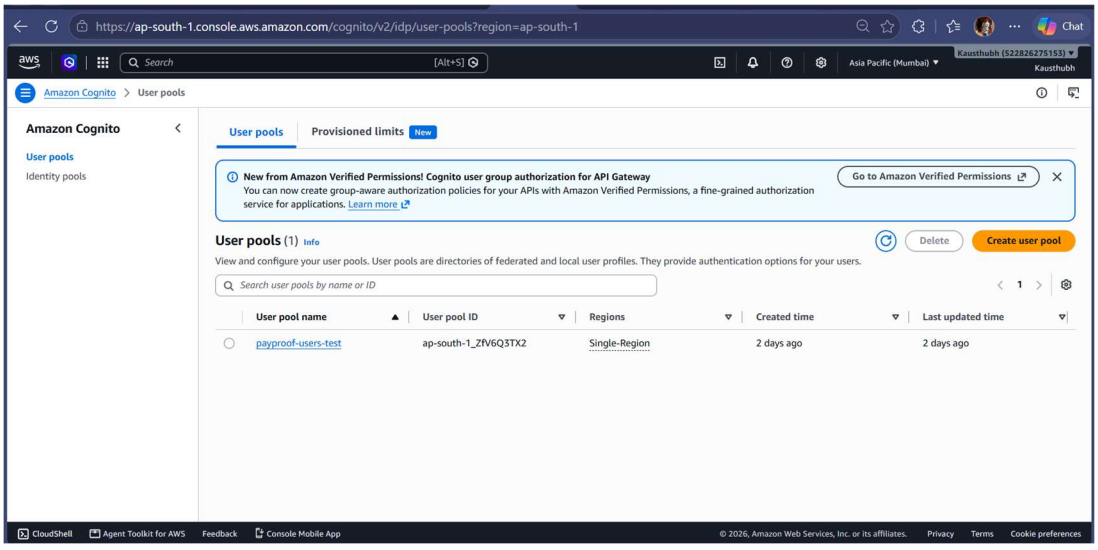


Figure 8.5 - Amazon Cognito User Pool inventory for PayProof.

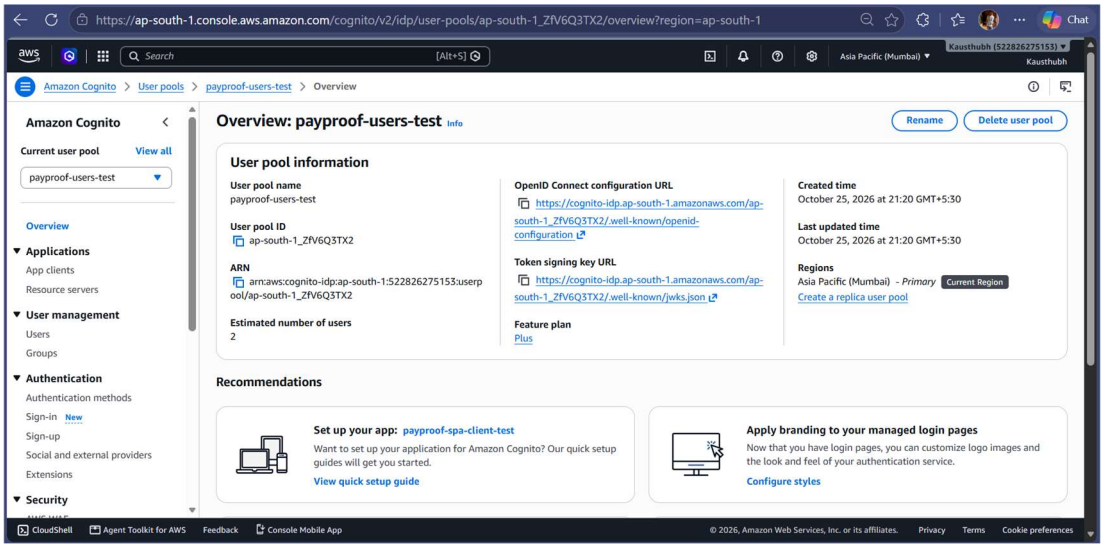


Figure 8.5(a) - Cognito User Pool overview for payproof-users-test.

8.6 Amazon S3 evidence bucket and stored objects

The S3 screenshots show the PayProof evidence bucket, an example evidence object with tenant/order key path, and a folder containing two PDF objects. They establish bucket/object presence only. The supplied views do not show Block Public Access, the bucket policy, default encryption, lifecycle/retention rules, or object-level permission outcomes.

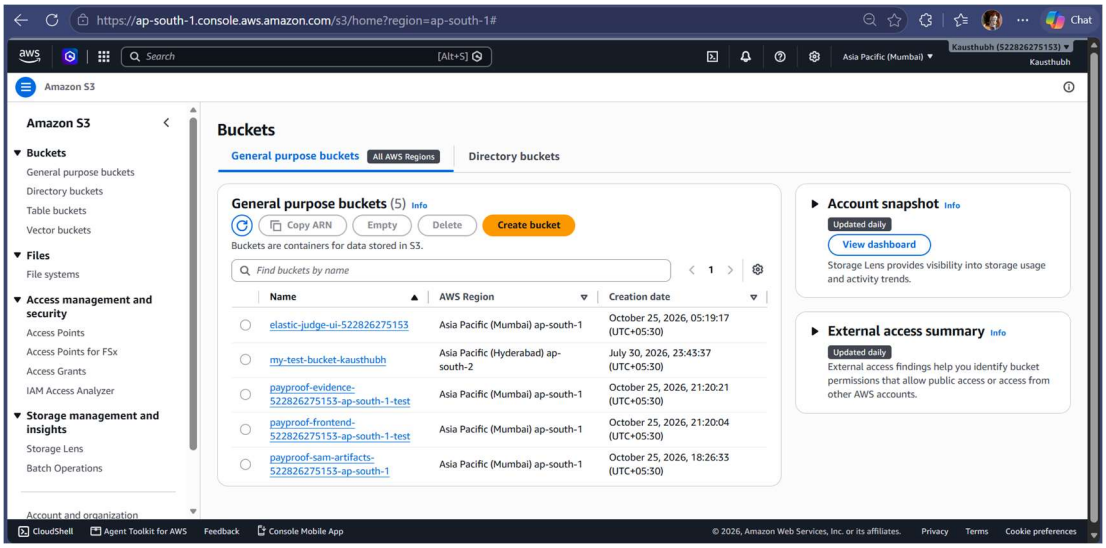


Figure 8.6 - Amazon S3 bucket inventory showing the PayProof evidence, frontend, and SAM artifact buckets.

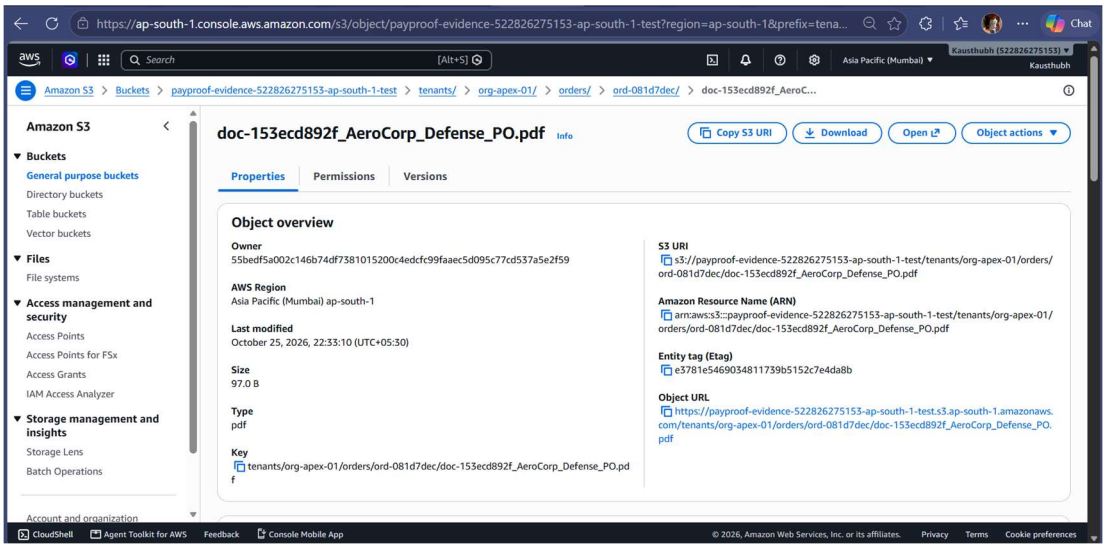


Figure 8.6(a) - S3 object properties for a sample stored order-evidence PDF.

8.7 S3 frontend bucket / deployment assets

The frontend bucket contains index.html, main.js, style.css, package.json, and vite.config.js. This confirms static assets are present; it does not establish the bucket access policy or whether CloudFront origin access control is correctly enforced.

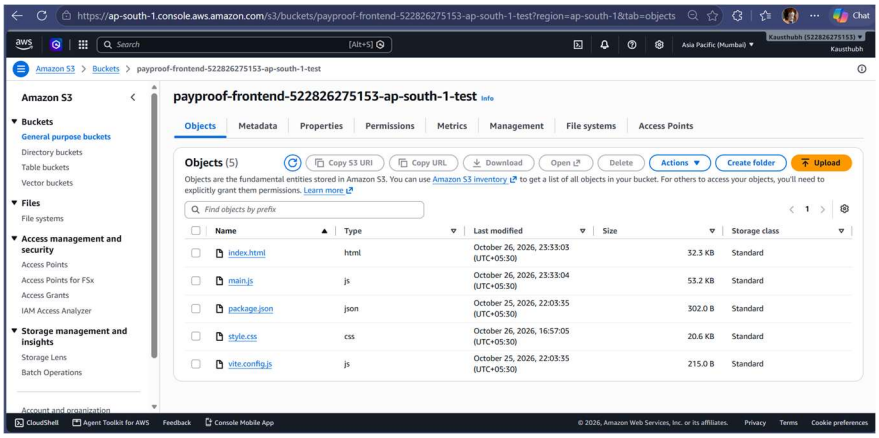


Figure 8.7 - PayProof frontend bucket contents, including the static website assets.

8.8 KMS customer-managed key

The KMS screenshots show a customer-managed key alias payproof-test with status Enabled, plus the broader customer-managed key inventory. The key details view includes part of the key policy. Confirm the full policy, rotation configuration, and actual use of this key by each target resource before asserting complete encryption coverage.

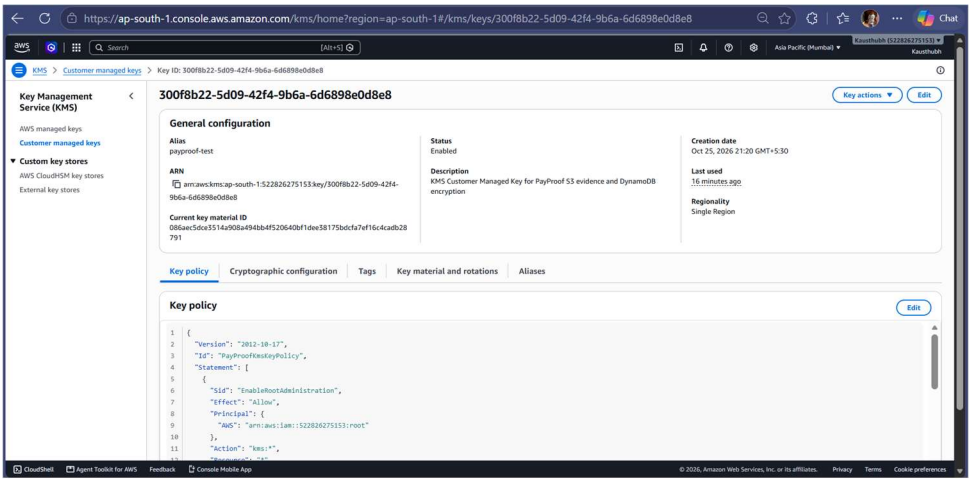


Figure 8.8 - AWS KMS key details for the enabled payproof-test customer-managed key.

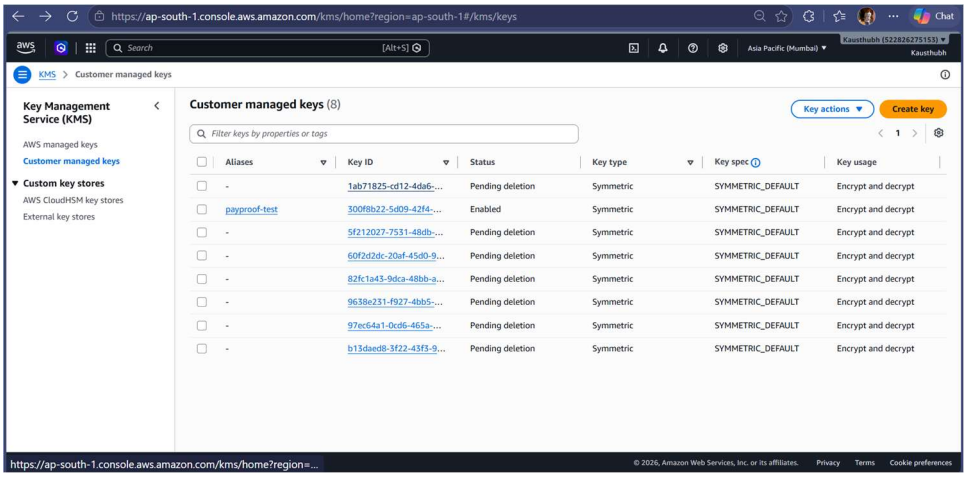


Figure 8.8(a) - KMS customer-managed key inventory showing the payproof-test alias enabled.

8.9 DynamoDB table and recovery settings

The DynamoDB console shows the payproof-records-test table as Active, with a String partition key (PK) and String sort key (SK), On-demand capacity mode, and point-in-time recovery (PITR) enabled. The visible item count is 0 and table size is 0 bytes at the time captured; this is a point-in-time observation and should not be interpreted as proof of historical or current production data volume.

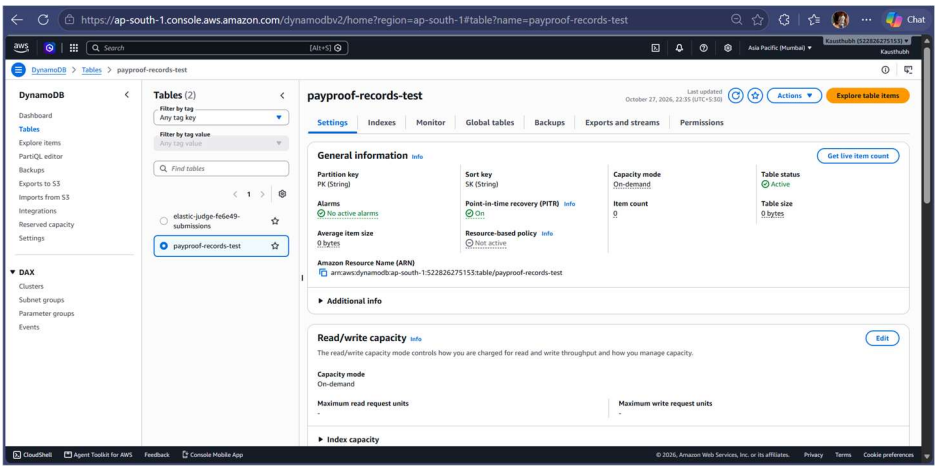


Figure 8.9 - DynamoDB payproof-records-test settings: key schema, On-demand capacity, Active status, and PITR enabled.

8.10 IAM Lambda execution roles and policy attachments

The supplied IAM console views show two Lambda execution roles and their attached policy names. PayProofApiExecutionRole-test lists AWSLambdaBasicExecutionRole, DynamoDBTableAccess, KmsDataKeyAccess, and S3EvidenceBucketAccess. PayProofOcrWorkerRole-test lists AWSLambdaBasicExecutionRole, S3AndDynamoDBReadWrite, SqsOcrProcessing, and TextractServiceAccess. These captures show attached policy names, not the full policy statements or trust relationship; least-privilege scope therefore remains to be reviewed.

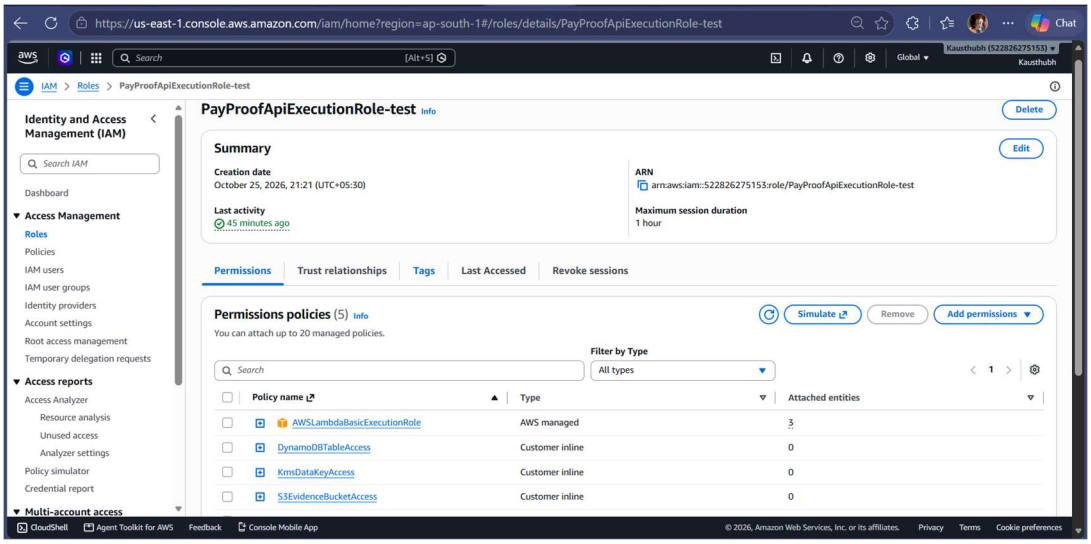


Figure 8.10 - PayProof API Lambda execution role and attached policy names.

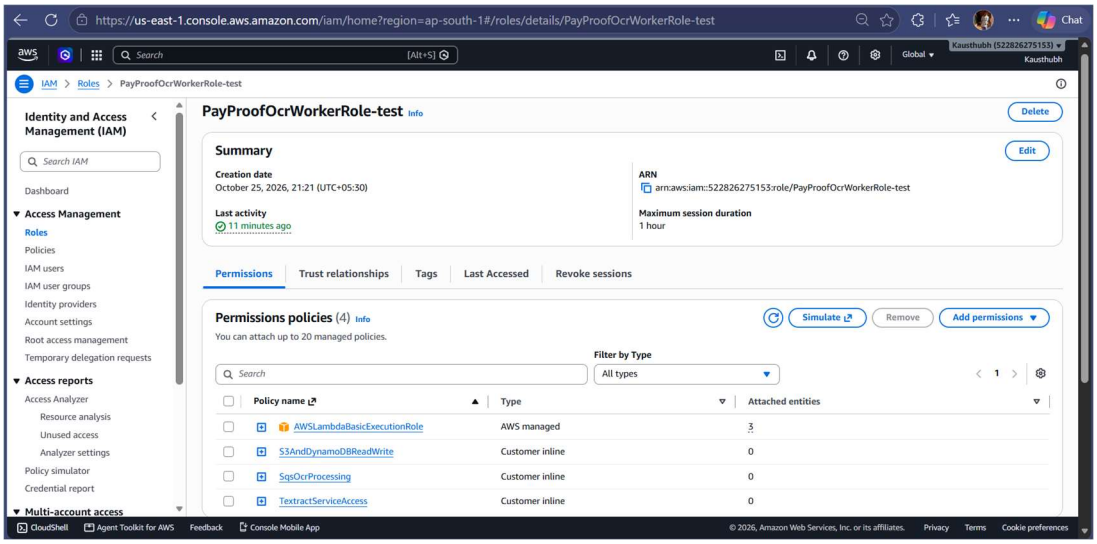


Figure 8.10(a) - PayProof OCR worker execution role and attached S3, DynamoDB, SQS, and Textract policy names.

8.11 CloudWatch API error metric

The CloudWatch screenshot displays the payproof-api-5xx-errors-test metric, with a visible datapoint at zero in the selected interval and a configured red threshold line. This does not establish longer-term reliability, log-group retention, or alarm action delivery. A CloudWatch Logs / retention screenshot was not supplied.

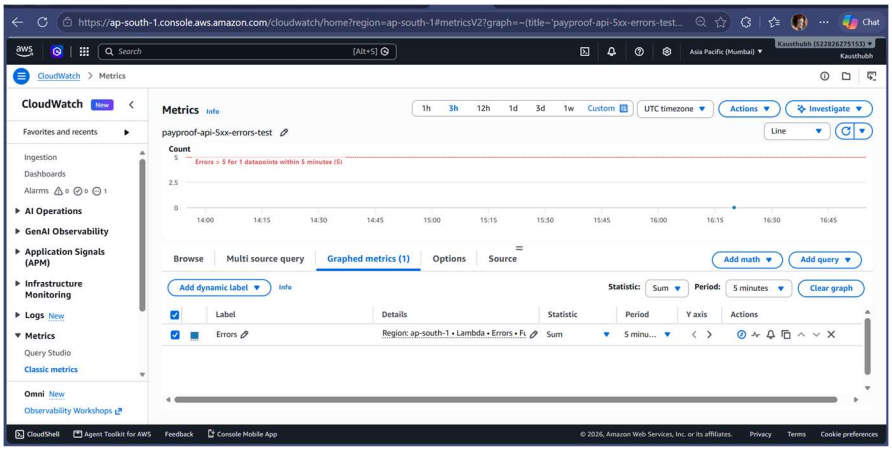


Figure 8.11 - CloudWatch metric view for the PayProof API 5xx-error metric.

8.12 Supplementary AWS Deployment Evidence

The following captures supplement Section 8. They document the stack listing and the presence of deployment artifacts in the SAM artifact bucket. They do not replace evidence of IAM permissions, bucket security, database recovery configuration, or a live application health check.

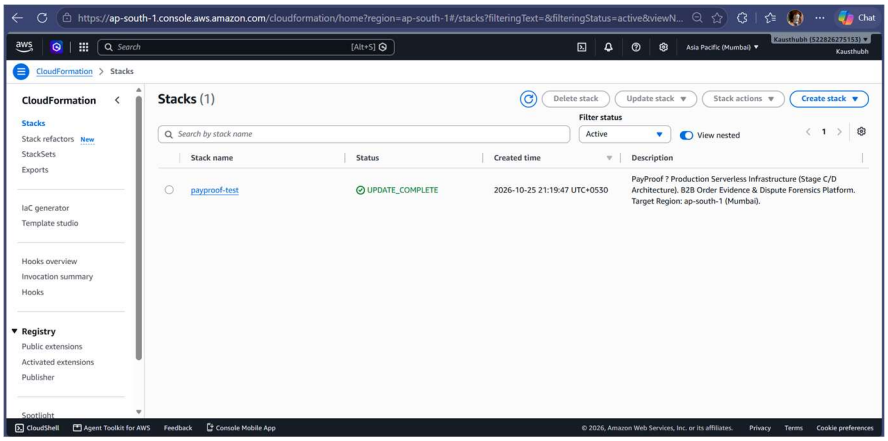


Figure 8.12 - CloudFormation stack inventory showing payproof-test in UPDATE_COMPLETE state.

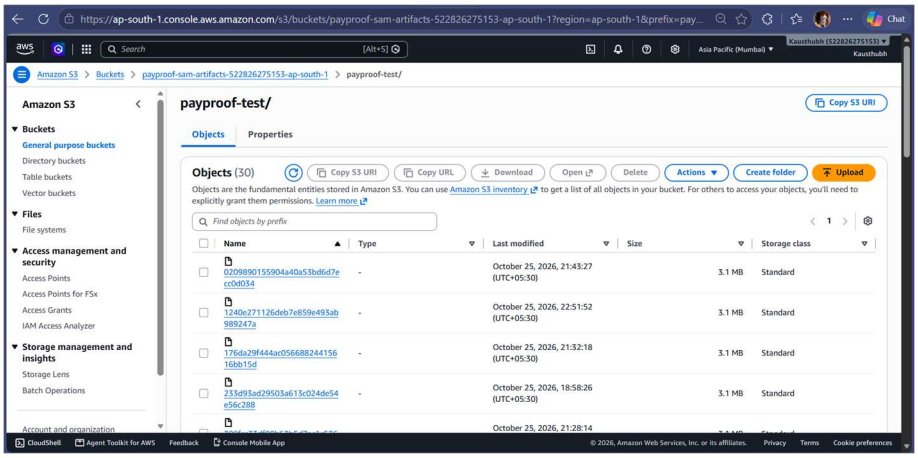


Figure 8.12(a) - SAM artifact bucket contents under the payproof-test deployment prefix.

9. Test Cases and Results

Project notes supplied for this report state that 85 local tests passed and that 14 deployment checks passed. They also report successful health checks (HTTP 200), unauthenticated API rejection (401), cross-tenant/spoofed access rejection (403), and order creation/retrieval checks. These results are recorded as owner-reported outcomes; this document author did not re-execute the suite or independently inspect live AWS state. Attach sanitized command output or test-run reports before final submission.

ID	Test case	Expected / procedure	Result and evidence status
TC-01	Frontend availability	Open reported CloudFront URL and check the application loads.	Reported successful; recheck live URL.
TC-02	Authentication required	Call a protected endpoint without a valid token.	Reported 401; attach sanitized response.
TC-03	Tenant authorization	Attempt an operation with a resource or organization identifier outside the user's scope.	Reported 403 for spoofed access; repeat across read, write, upload, and export.
TC-04	Order creation and retrieval	Create an order and read it back after reload.	Reported pass; provide persistence evidence.
TC-05	Evidence upload	Upload a supported evidence file and verify object/metadata persistence.	Pending independent verification.
TC-06	Evidence hash validation	Recompute SHA-256 from a downloaded original and compare against stored/exported metadata.	Pending independent verification.
TC-07	Lifecycle enforcement	Attempt valid and invalid transitions; test override permissions.	UI warning visible; server enforcement needs test evidence.
TC-08	Evidence-pack export	Inspect ZIP entries, manifest, hashes, and organization boundaries.	Download UI visible; archive validation pending.
TC-09	Dispute resolution	Update a dispute and reload to confirm persisted resolution/history.	UI flow visible; persistence test evidence needed.
TC-10	Audit-log protections	Attempt unauthorized audit-event alteration/deletion and verify override events.	Pending AWS policy and negative-test evidence.
TC-11	Health check	Call the deployed health endpoint and inspect dependency status.	HTTP 200 reported; include sanitized output.
TC-12	Automated/local regression suite	Execute the repository test suite.	85 local tests reported passing; attach dated test output.

10. Risks and Recommendations

Risk area	Recommendation
Authentication and demo access	Rotate any previously shared demo credential and remove public one-click demo access before real customer data is handled. Confirm token expiry and sign-out behavior.
Tenant isolation	Enforce organization/tenant checks server-side on every order, evidence object, export, dispute, and audit endpoint. Test cross-tenant negative cases by changing client-supplied identifiers.
File uploads and malware	Treat new uploads as untrusted. Validate size/type server-side, use safe object keys, quarantine files, and confirm a real malware scan verdict before use. A PENDING_SCAN label alone is not evidence of scanning.
Evidence integrity and export	Document the hash format and calculation path. Independently verify ZIP structure, manifest-to-file correspondence, and per-file digests. Clarify whether hashes are stored and whether exports are signed.
Audit immutability	Restrict update/delete permissions for audit records and verify the controls with negative tests. Use retention/versioning or an append-only design if tamper resistance is a requirement.
Encryption and key access	Verify evidence bucket privacy and encryption. If KMS is used, review key policy, grants, rotation, and least-privilege use; do not claim KMS encryption unless configuration proves it.
Monitoring and recovery	Set log retention intentionally, redact tokens and document content from logs, configure alarms for errors and access anomalies, enable appropriate backups, and test restoration.
Production readiness	Run a controlled pilot using synthetic or approved data; document privacy, retention/deletion, support, incident handling, and operational ownership before onboarding real customers.

11. Conclusion

PayProof demonstrates a substantial MVP interface for B2B order evidence and reconciliation. The supplied screenshots show order dashboards, order creation, evidence ingestion, completeness indicators, lifecycle warnings, evidence-pack export, dispute management, an audit-ledger display, evidence-integrity guidance, and a Cognito sign-in interface. Project notes further describe a test deployment using AWS-managed frontend, API, identity, compute, object-storage, key-management, database, and monitoring services.

The report intentionally separates demonstration evidence from production guarantees. Live AWS configuration, access-control policies, data durability, malware scanning, hash verification, audit protections, backup/recovery, and operational monitoring should be verified with the real screenshots and tests listed in Section 8 and Section 9. On the evidence presently available, the status is: MVP demonstrated; pilot readiness pending verification. Production-ready SaaS status should not be claimed until the outstanding controls have been demonstrated and documented.

12. References

[1] Amazon Web Services, “Amazon CloudFront Developer Guide.” https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html

[2] Amazon Web Services, “Amazon API Gateway Developer Guide.” https://docs.aws.amazon.com/apigateway/latest/developerguide/welcome.html

[3] Amazon Web Services, “AWS Lambda Developer Guide.” https://docs.aws.amazon.com/lambda/latest/dg/welcome.html

[4] Amazon Web Services, “Amazon Cognito Developer Guide.” https://docs.aws.amazon.com/cognito/latest/developerguide/cognito-user-identity-pools.html

[5] Amazon Web Services, “Amazon S3 User Guide.” https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html

[6] Amazon Web Services, “Amazon DynamoDB Developer Guide.” https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Introduction.html